GATE

Computer Science & Information Technology

Discrete Mathematics

DPP: 6

Set Theory and Algebra

Q1 The set of all positive rational numbers forms an abelian group under the composition * defined by a * b = (ab)/2.
Which of the following is/are TRUE?
(A) The identity element is 2
(B) The inverse of a is 4/a.
(C) The inverse of 4 is 1
(D) The identity element is 1

Q2 Let r be the set of all real numbers and * is a binary operation defined by a * b = a + b + ab.
Which of the following is TRUE?
(A) Identity element is 0.
(B) The inverse of –1 is 1.
(C) The inverse of a is –a/(a + 1).
(D) R is not a group.

Q3 The set G = {0, 1, 2, 3, 4, 5} is a group with respect to addition modulo 6.
Which of the following is false?
(A) The inverse of 2 is 4
(B) The inverse of 3 is 3
(C) The inverse of 5 is 2
(D) The inverse of 1 is 5

Q4 G = {1, –1, i, –i} is a group with respect to multiplication. The order of –i is

Q5 If G is a group of order p, where p is a prime number. Then the number of sub groups of G is____.
(A) 1 (B) 2
(C) p –1 (D) p

Q6 Which of the following is/are True?
(A) The union of any two sub groups of a group G is also a sub group of G.
(B) The intersection of any two sub groups of a groups of a group G is also a sub group of G.
(C) Let G = {0, ± k, ± 2k, ±3k, ±4k, …..∞} where k is any fixed integer is a group w.r.t. addition.
(D) Every sub group an abelian group is also an abelian group.

Q7 Which of the following is/are True?
(A) Every cyclic group is an abelian group
(B) If 'a' is generator of a cyclic group then a^{-1} is also a generator of G.
(C) The order of a cyclic group is equal to the order of its generating element.
(D) A sub group of a cyclic group need not be cyclic.

Q8 How many generators are there for the cyclic group ({0,1, 2, 3, 4, 5, 6}, $\oplus_7$)

Q9 Which of the following is false?
(A) A cyclic group with only one generator can have at most 2 elements.
(B) The order of a cyclic group is equal to the order of its generator.
(C) The group ({1, 2, 3, 4}, $\otimes_5$) is cyclic
(D) A group of order 4 is cyclic.

Q10 Let S = {0, 1, 2, 3, 4, 5, 6, 7} and * denotes multiplication modulo 8 that is x * y = (xy) mod 8.
Which of the following is a group with respect to *?
(A) {0, 1} (B) {1, 4}
(C) {1, 3} (D) {1, 6}

Answer Key

Q1 **(A, B, C)**

Q2 **(A, C, D)**

Q3	(C)
Q4	4 to 4
Q5	(B)
Q6	(B, C, D)
Q7	(A, B, C)
Q8	6
Q9	(D)
Q10	(C)

GATE

Hints & Solutions

Q1 Text Solution:

Let e be the identity element.

$\therefore$ a * e = a

$\Rightarrow$ (ae/2) = a

$\Rightarrow$ e = 2

$\therefore$ Option (a) is true and option (d) is false.

Let a^{-1} = inverse of a

a * a^{-1} = e

$\Rightarrow \frac{a \times a^{-1}}{2} = 2$

$\Rightarrow a^{-1} = \frac{4}{a}$

Inverse of $4 = \frac{4}{4}$

$\therefore$ Option (b) and (c) are true.

Q2 Text Solution:

Let e be the identity element.

$\therefore$ a * e = a

$\Rightarrow$ a + e + a. e = a

$\Rightarrow$ e = 0

Let a^{-1} = inverse of a

a * a^{-1} = e

a + a^{-1} + aa^{-1} = 0 (\ 0 is identity element)

$\Rightarrow a^{-1} = \frac{-a}{a+1}$

$\therefore$ Inverse of –1 does not exist.

Hence, Option (b) is false.

Q3 Text Solution:

The identity element is 0.

$5 \oplus_6 2 = 1$

$\Rightarrow$Inverse of 5 is not 2.

Q4 Text Solution:

Order of (–i) = 4, because the smallest integer n such that $(-i)^n$ = 1 (identity element) is n = 4.

Q5 Text Solution:

Let (H, *) be a subgroup of order n, By Lagrange's theorem,

$\Rightarrow$ n is a divisor of p

$\Rightarrow$ n = 1 or n = p

$\Rightarrow$ H = {e} or H = G

$\therefore$ G has only 2 trivial subgroups

Q6 Text Solution:

G = {1, 3, 5, 7} is a group with respect to $\otimes_8$.

H_1 = {1, 3} and H_2 = {1, 5}

$H_1 \cup H_2$ = {1, 3, 5}

Here, H_1 and H_2 are subgroup of G,

But $H_1 \cup H_2$ is not a subgroup of G.

Q7 Text Solution:

Every sub group of a cycle group is cyclic (theorem).

Q8 Text Solution:

Number of generator of G = ϕ(7) = 7-1 = 6

Q9 Text Solution:

The Klein Four Group V_4 (also known as $\mathbb{Z}_2 \times \mathbb{Z}_2$):

- This group consists of four elements: $e, a, b,$and ab, where:
  - $a^2 = e, b^2 = e,$ and $(ab)^2 = e,$
  - The operation between any two elements results in another element within the set.
- This group is **not cyclic** because there is no single element that can generate all elements of the group. Instead, it is an abelian group with each non-identity element having order 2.

Hence, option D is false.

Q10 Text Solution:

(a) {0, 1} is not a group, because inverse of 0 does not exist.

(b) {1, 4} is not a group, because the set is not closed with respect to the given binary operation.

(c) {1, 3} is closed with respect to *.

$\therefore$ {1, 3} is a group.

(d) {1, 6} is not a group, because the set is not closed with respect to the given binary operation.

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